Lab 32: Plan first, then approve into an editing mode

Topic 3: Claude Code Configuration & Workflows · Task 3.4: Determine when to use plan mode vs direct execution

Runs on Simulator 2: Code Generation with Claude Code · Self-practice

Source: SkillCertPro practice exam Part 1, Q30 (unofficial third-party question)

The question

A senior engineer is about to migrate a shared authorization layer used by several packages. The dependency graph is unclear, there are multiple viable designs, and prior immediate-edit attempts caused broad rewrites that were later reverted. What should the engineer do first?

A. Switch to bypass-permissions mode, let Claude inspect and edit freely, then rely on tests to catch bad changes.

B. Use direct execution, ask Claude to update one package at a time, then revert commits if the design shifts.

C. Start Claude Code in plan mode, have it inspect the affected areas, review the proposed plan, then approve into an editing mode.

D. Start Claude Code in accept-edits mode, have it inspect the affected areas, then review the resulting source changes.

Simulate the solution

```
# In the browser: open Simulator 2, then Lab 32
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 32

# In a terminal, from simulators/simulator-2-code-generation/
uv run casekit.py --case UC-30
```

What you should see

Plan phase changes nothing. The same session resumes with acceptEdits, and the edits land.

Phase 1 (plan mode) changes nothing. Phase 2 resumes the same session with acceptEdits and the approved first step lands as real edits.

Open `simulators/simulator-2-code-generation/evidence/` and find it yourself.

The answer

C. Plan mode, review the plan, approve, then edit.

Plan mode is for situations where changing code immediately is risky because the design, dependencies or scope are unclear. Claude analyses the codebase and writes a plan, but source files are not edited until the plan is approved. Approving the plan exits plan mode into an editing mode, either reviewing each edit or accepting edits. Don't confuse plan mode with the more permissive modes: those govern approval behaviour, not whether analysis comes first.

Why the others fail

A. The most permissive mode repeats the earlier broad-rewrite failure.

B. Commits to a design before one has been chosen, then pays for it in reverts.

D. Accept-edits approves edits automatically, so review happens only after the changes are made.

Check yourself

- plan mode inspected without editing
- approval continued the same session (resume, not a new chat)
- the approved step produced edits (a new or changed source or test file)

Now change one thing

Set LAB_MODEL=claude-sonnet-5-5, rerun, and compare the result and the cost line in evidence/.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.